

ADITYA RAI

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PROFESSIONAL SUMMARY

AI and Machine Learning engineer with hands-on industry experience in computer vision, deep learning, and anomaly detection at the Indian Institute of Science. Proficient in building end-to-end ML pipelines, optimizing neural network models, and deploying AI tools for real-world applications. Pursuing B.Tech in Mathematics and Computing (3rd year). Seeking AI/ML/Computer Vision internship roles at leading technology organizations.

TECHNICAL SKILLS

Programming Languages: Python, C, C++

ML Frameworks & Libraries: PyTorch, YOLOv8, DenseNet121, ResNet, CNNs, PaDiM

Computer Vision: OpenCV, Image Classification, Object Detection, Anomaly Detection

Data Science Tools: pandas, NumPy, SQL, Gradio, Data Visualization, Dataset Preprocessing

Concepts: Deep Learning, Transfer Learning, Model Training & Evaluation, Data Augmentation, Statistical Analysis

EXPERIENCE

AI Intern – Railway Safety

Jun 2025 – Aug 2025

Lab to Market Program, Indian Institute of Science (IISc)

Bangalore, India

- Diagnosed operational gaps in railway inspection processes and designed data-driven solutions using ML-based anomaly detection to automate defect identification.
- Developed and evaluated anomaly detection models using PaDiM and deep learning techniques, improving inspection reliability and reducing manual error rates.
- Conducted structured performance benchmarking (precision, recall, F1-score, AUC-ROC) to compare model outputs and inform engineering improvements.
- Collaborated with IISc researchers and cross-functional teams to translate ML workflows into clear technical documentation and deployment-ready insights.

PROJECTS

Chest X-Ray Analysis for Early Disease Detection | *PyTorch, DenseNet121, OpenCV, Gradio* | [GitHub](#) 2025

- Built an end-to-end deep learning pipeline for multi-label chest X-ray abnormality detection using DenseNet121 with transfer learning and fine-tuning.
- Achieved **87.8% classification accuracy** through model architecture optimization, class-weighted loss functions, and targeted data augmentation strategies.
- Improved dataset reliability by implementing automated quality checks to detect and filter mislabeled or corrupted images from training data.
- Deployed an interactive Gradio interface providing real-time prediction confidence scores and diagnostic insights for clinical review.

EDUCATION

Madhav Institute of Technology and Science (MITS)

2023 – 2027

B.Tech in Mathematics and Computing

Gwalior, India

DAV ACC Public School

2008 – 2022

Senior Secondary – Physics, Chemistry, Mathematics (PCM)

Kymore, India

ACHIEVEMENTS

KBG Hackathon 2026 – Top 30 Finalist

2025

Kamand Bioengineering Group, IIT Mandi

- Selected among top 30 teams out of 1000+ registrations based on mid-submission evaluation; presented on-site at IIT Mandi competing against shortlisted teams nationwide.
- Built a Hybrid CNN + ViT bone fracture classifier (EfficientNet-B3 + ViT-Small via cross-attention fusion) achieving 98.22% test accuracy and 0.997 AUC-ROC on X-ray radiographs.

Smart India Hackathon – National Finalist

2024

Government of India

- Selected among top teams nationally; recognized for structured AI-driven problem-solving, rapid prototyping, and technical solution design under competitive time constraints.